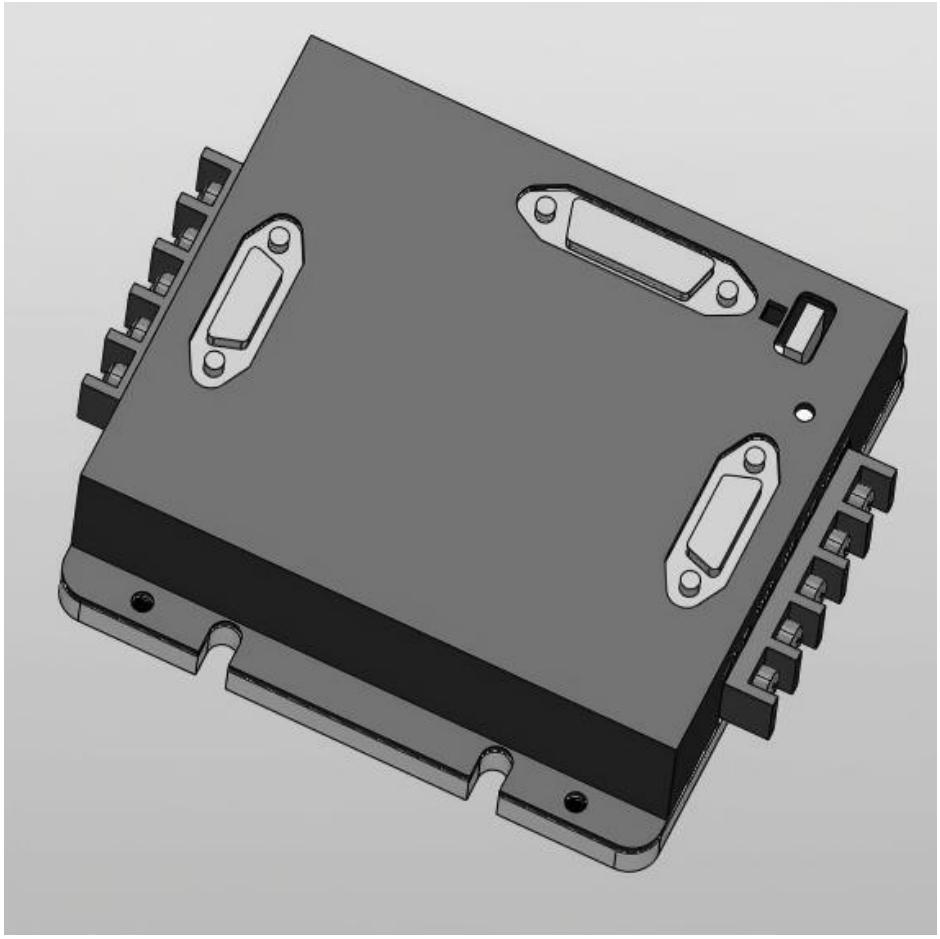
LKBS48302C
Dual-channel brushless servo driver operating instruction



Company: Jinan Lanjiu Electronic Technology Co., Ltd.

Address: Room 101, Building 25, Liandong U Valley, Yuqing Road, Changqing District, Jinan City, Shandong Province, China 250300

Contact Person: Denise / Ms. Zhang

Tel: +86-18340079571

Email:wangxiao@lanjiukeji.com

Version V1.0

Catalogue

I. Performance Description-3- ..... - 3 - 2

2. Product dimensions: (115 * 95 * 40mm) -4- ..... - 4 - 3

3. Wiring Diagram and Interface Definitions-5- ..... - 5 - 4

3.1. Battery wiring diagram: The driver has a power connector on each side (left and right), allowing connection to either side as shown in the figure: -5- ..... - 5 - 5

3.2. Motor Wiring Diagram-5- ..... - 5 - 5

3.3. Definition of Power Section Wiring-6- ..... - 6 - 5

3.4. Control section wiring definition (P1) (Note: 3 and 5 are not enabled) -6- .- 6 - 6

3.5. Wiring specifications for the feedback signal section (P2 and P3) -7- ..... - 7 - 7

IV. CAN Communication Specifications-8- ..... - 8 - 8

1. CAN Configuration-8- ..... - 8 - 8

2. Control Instructions-9- ..... - 9 - 9

3. Query command-10- ..... - 10 - 10

5. LED Indicator Status Description-12- ..... - 12 - 12

1、performance specification

- Input voltage: 12-48VDC, with overvoltage limit at 60V and undervoltage limit at 10V.
- Output current: continuous 2×15A, peak 2×30A (10S);
- Work mode: Standalone mode, Hybrid mode;
- operate mode ：

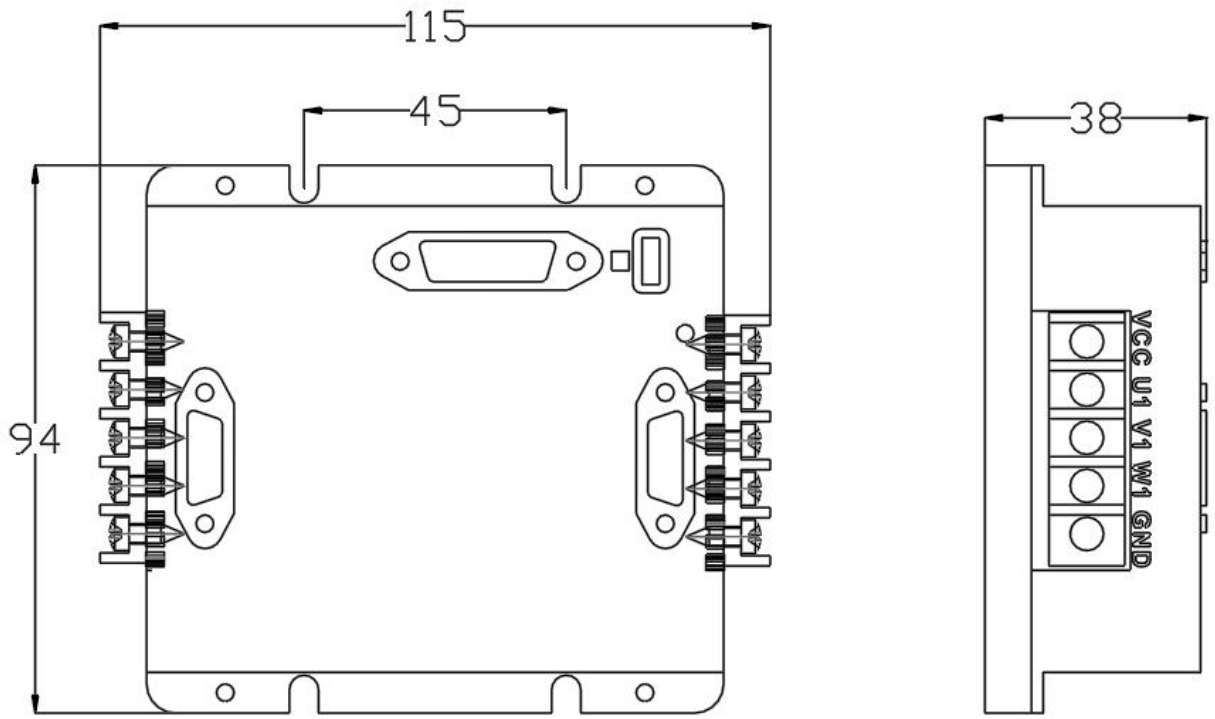
  Open-loop speed control and closed-loop speed control (using incremental encoder);
- control method ：

  Analog input: 0-5V/0-2.5-5V, with a joystick set to 2.5V as the default value.

  Model aircraft signal: 1000-1500-2000us;

  PWM signal: 0-99% (recommended 250Hz, 5V amplitude);
- Communication methods: RS232, CAN;
- Protection modes: overcurrent, overvoltage, undervoltage, overheating, and short circuit;
- Operating environment: relative humidity ≤80%, no condensation, temperature range-20 to 60℃;
- Weight: approximately? kg;

2、Product dimensions: (115 * 95 * 40mm)



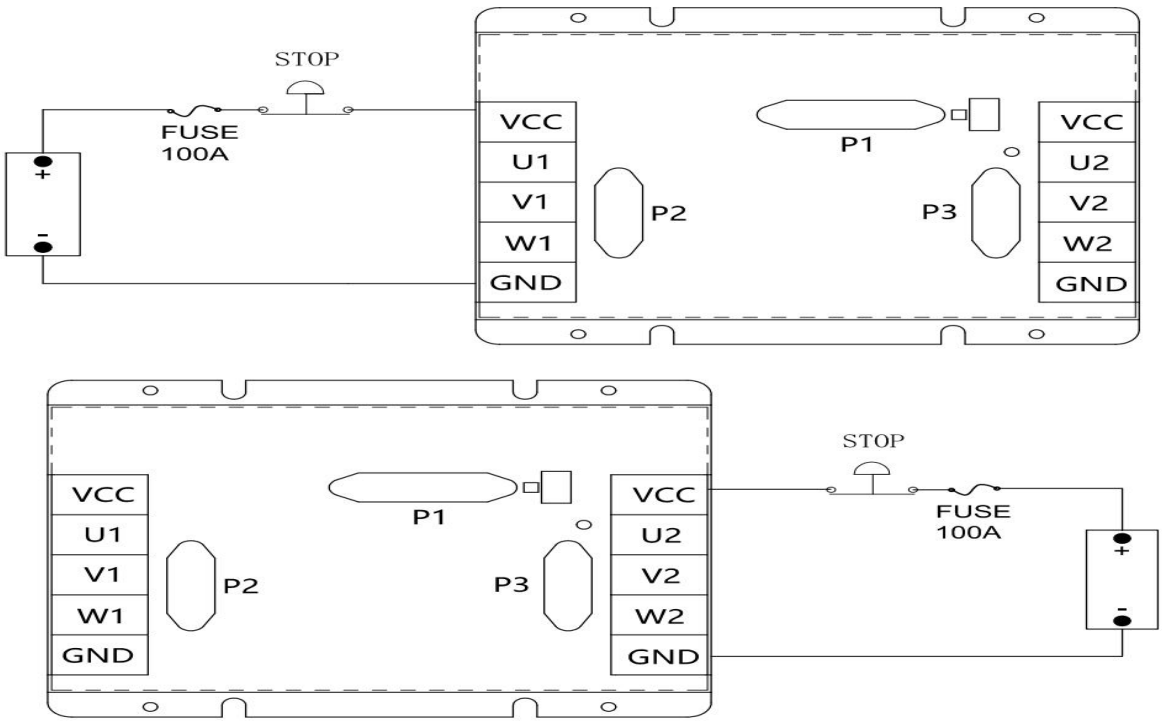
pour ：

1. The drive unit is equipped with four Φ5mm mounting and fixing holes. During installation, apply thermal grease to the bottom of the radiator and secure it tightly against the machine housing to facilitate heat dissipation. The operating environment should be free from dust, steam, flammable gases, corrosive gases, and oil mist. Additionally, anti-vibration measures should be implemented.

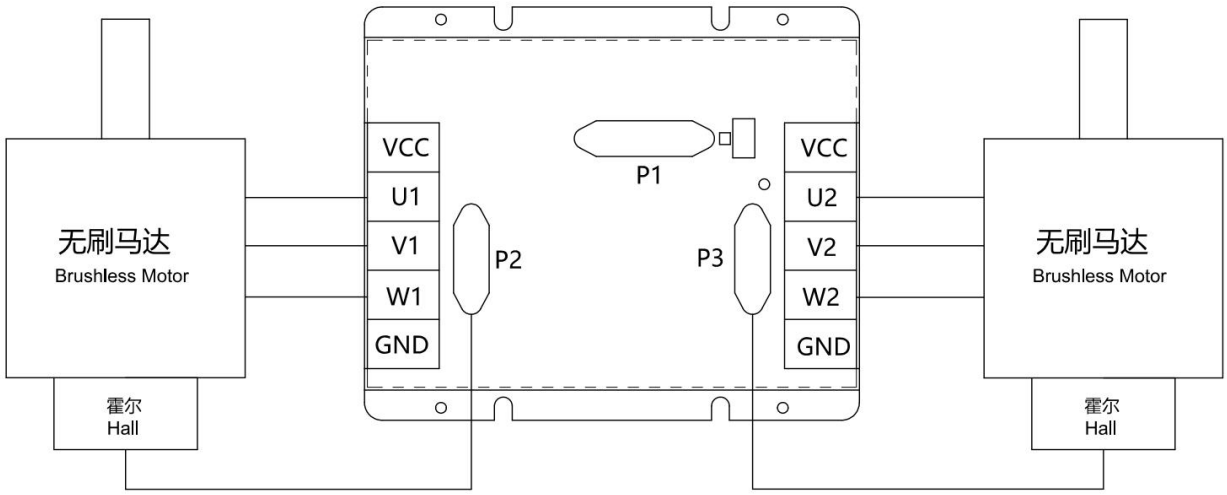
2. Leave a 5-8cm clearance at the top of the drive (for control and signal connectors).

3、Wiring diagram and interface definition

3.1. Battery wiring diagram: The driver has a power connector on each side (left and right), allowing connection to either side as shown in the figure.



3.2 Motor Wiring Diagram

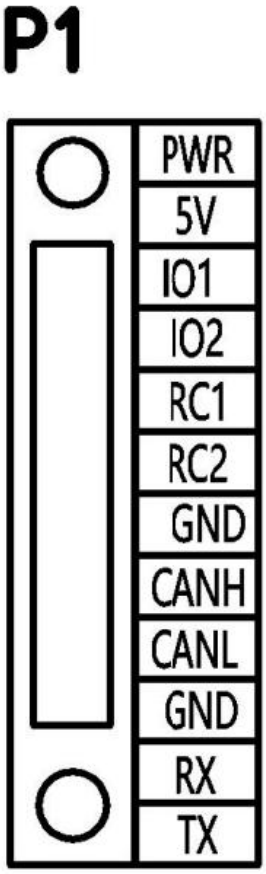


3.3. Definition of Power Section Wiring

port	explain
VCC	positive terminal of power supply
GND	negative terminal of power supply
U	Motor U-phase
V	Motor V-phase
W	Motor W-phase

The drive has no anti-reverse function. Ensure the polarity of the external power supply matches the drive's markings exactly, or it may cause permanent damage.

3.4. Control section wiring definition (P1) (Note: 3 and 5 are not enabled)



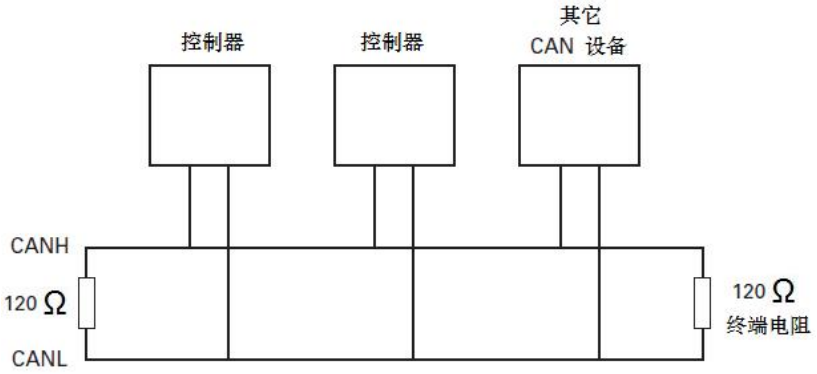
port	definition	function
1	CAN_L	CAN_L
2	TX	RS232-TX
4	GND	Signal ground (GND)
6	+5V	5V power supply
7	+5V	5V power supply
8	PWR	Control power input, reserved for customization
9	CAN_H	CAN_H
10	RX	RS232-RX
11	GND	Signal ground (GND)
12	RC2	Signal Input Port 2
13	RC1	Signal Input Port 1
14	IO2	obligate
15	IO1	obligate

- Reserved ports IO1 and IO2:

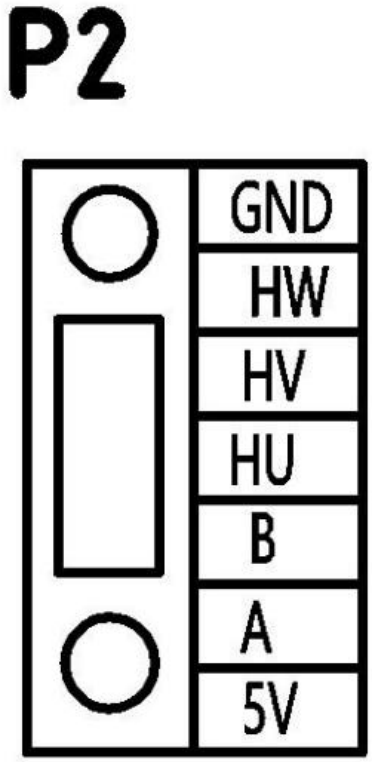
By default, the factory configuration connects the direction control terminals of Motor 1 and Motor 2 to +5V to switch the motor's rotation direction. Alternatively, it can be configured for 0-2.5-5V (joystick) control based on customer requirements.

- CAN port:

Note: The drive contains a built-in 120-ohm resistor. Select it via the DIP switch by pressing the 120Ω series resistor.



3.5. Wiring specifications for the feedback signal section (P2 and P3)



port	definition	function
1	+5V	5V power supply
2	HA	Hall signal A
3	HB	Hall signal B
4	HC	Hall signal C
5	GND	Signal ground (GND)
6	+5V	5V power supply
7	A	Encoder signal A
8	B	encoder signal
9	GND	Signal ground (GND)

4、CAN Communication Specification

1. CAN configuration

(1) Frame format: Standard frame

(2) CAN type: CAN2.0A

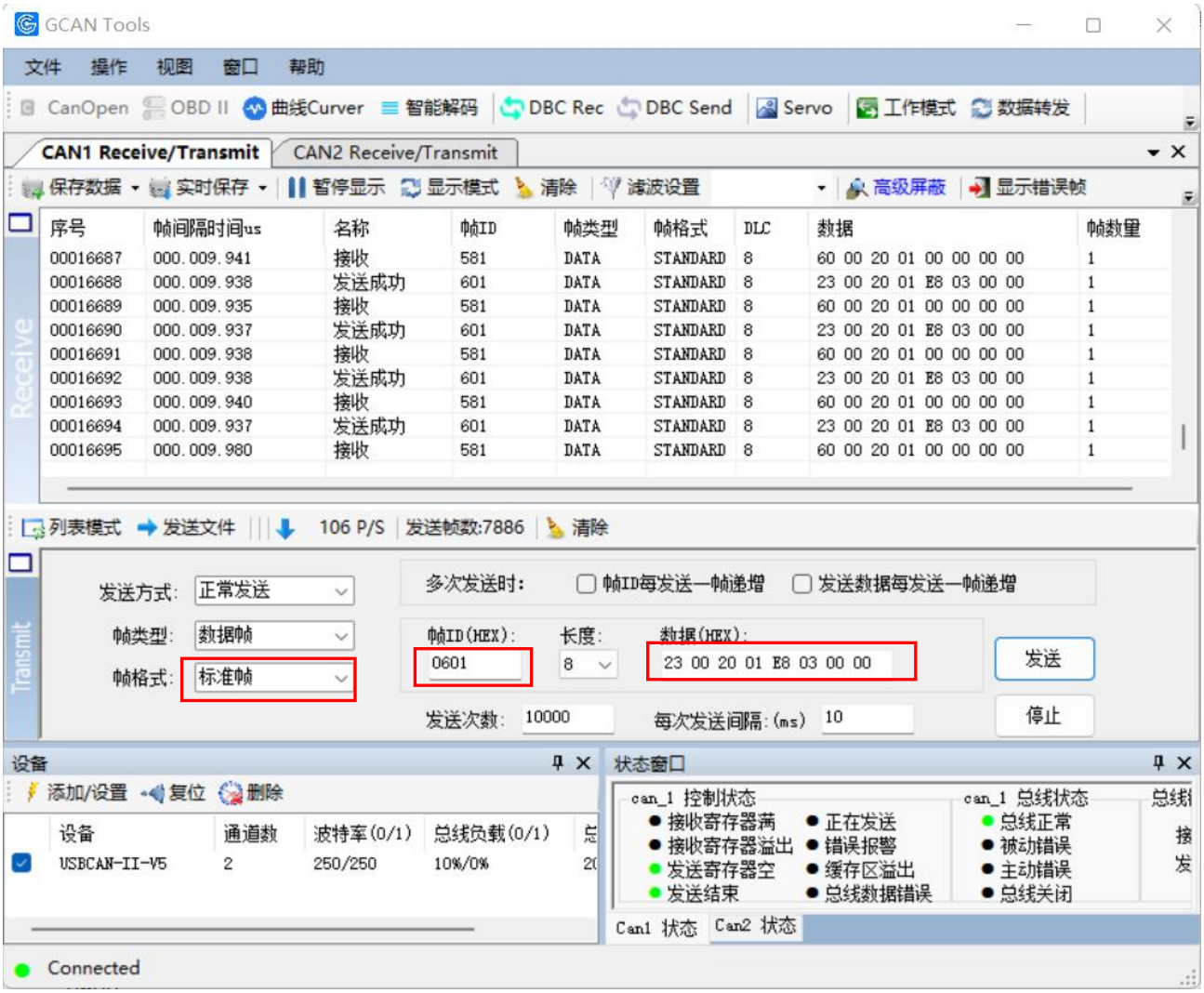
(3)Baud rate: 125k; 250k (default); 500k;

(4) Node ID: 1 (The ID can be configured via the host computer software, with the default value being 1 by default);

(5) Disconnection protection duration: 1000ms (during which no data is received, triggering the protection);

(6) Auto-Start: Enabled by default;

(7) Data format: lower bits first, higher bits last;



2. Control Instructions
Send command:
ID: 0x600+ target address ID (0x01-0x7E)

Data_0	Data_1、2	Data_3	Data_4	Data_5	Data_6	Data_7
0x23	index of matrix	subindex	XX	XX	XX	XX

bridging order ：

ID: 0x580+ Target address ID (0x01-0x7E)

Data_0	Data_1、2	Data_3	Data_4	Data_5	Data_6	Data_7
0x60	index of matrix	subindex	00	00	00	00

Note: See the appendix for indexes and sub-indexes.
Note: The drive is enabled by default after startup. You can run the motor by sending control commands directly. If the motor stops during operation, you need to send an enable command or restart to reset.
Example: ID 1 controller, open-loop mode, index 0x2000, setpoint range (-1000 to 1000).
Send ID: 0x601
Return ID: 0x581
The first channel is set to 1000 (0x03E8), with full forward voltage output (95% input voltage).
Send data: 0x23 00 20 01 E8 03 00 00
Return data: 0x60 00 20 01 00 00 00 00

The second channel is set to-500 (0xFFFE0C, i.e., the complement of 0x01F4 plus 1), with reverse half-output voltage.
Send data: 0x23 00 20 02 0C FE 00 00
Return data: 0x60 00 20 02 00 00 00 00

disability ：

Send data: 0x2F 0C 20 00 00 00 00 00
Return data: 0x60 0C 20 00 00 00 00 00

enable ：

Send data: 0x2F 0D 20 00 00 00 00 00
Return data: 0x60 0D 20 00 00 00 00 00

3. Query command
Battery voltage query (0x210D):
Send: 0x40 0D 21 02 00 00 00 00
Return: 0x4B 0D 21 02 EF 00 00 00
0xEF=239, battery voltage=23.9V.

Current query (0x2100) for channel 0x01 and channel 0x02:
Send: 0x40 00 21 01 00 00 00 00
Return: 0x4B 00 21 01 10 00 00 00
0x10=16, motor current=1.6A.

Temperature query (0x210F) for channel 0x02 and channel 0x03:
Send: 0x40 0F 21 02 00 00 00 00
Return: 0x4B 0F 21 02 1C 00 00 00

0x1C=28， temperature =28℃。

(Loop) Encoder speed query (0x2103) for channel 0x01 and channel 0x02:
Send: 0x40 03 21 01 00 00 00 00
Return: 0x4B 03 21 01 C8 00 00 00
0xC8=200, encoder speed=200 RPM.

Fault query (0x2112):
Send: 0x40 12 21 00 00 00 00 00
Return: 0x4B 12 21 00 02 00 00 00
0x02=2, see fault status table, overvoltage/undervoltage fault.

matters need attention ：

1. "XX" indicates a valid byte of data. "00" indicates an invalid byte with no defined data.
2. Both the transmitting and receiving data fields in CAN protocol are 8 bytes, with any extra bytes being invalid.
3. In multi-byte indexes and data structures, the low byte is placed before the high byte.
4. Invalid commands will result in error frames, and such knowledge is not covered by this agreement.

5、LED indicator status description

Normal state (POWER green light on, STATUS red light indicating the given signal mode)	
Status indication (period 1S)	mode declaration
STATUS red light flashes once	digital signal input mode
STATUS red light flashes twice	pulse input mode
STATUS red light flashes three times	analog input mode

Fault status (POWER green light on continuously, STATUS red light flashing to indicate fault)	
Status indication (period 1S)	Fault Description
STATUS The red light flashes rapidly	short
STATUS Flash red light for 4 seconds, then stay on for 1 second	Under-voltage or over-voltage
STATUS Flash red light for 2 seconds, then stay on for 1.5 seconds	superheat